

Statistical and Machine Learning

Evaluation of model performance

Proof sketch: Cook's distance and the F distribution

Consider the linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, where $\mathbf{X}$ is $n \times p$. Cook's distance for observation i is defined as

$$D_i = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})}{p \hat{\sigma}^2},$$

where $\hat{\boldsymbol{\beta}}_{(i)}$ is the OLS estimate with observation i deleted.

Step 1 — Sherman–Morrison update.

Applying the Sherman–Morrison formula to the rank-1 change in $\mathbf{X}^\top \mathbf{X}$ when row i is removed gives

$$\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = \frac{e_i}{1 - h_{ii}} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i,$$

where $e_i = y_i - \hat{y}_i$ is the i -th OLS residual and $h_{ii} = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i$ is the i -th leverage (diagonal entry of the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$).

Step 2 — Substituting into D_i .

Plugging the update from Step 1 into the definition of D_i and using $\mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i = h_{ii}$:

$$D_i = \frac{e_i^2}{p \hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}.$$

Step 3 — Studentized residual form.

Define the internally studentized residual

$$r_i = \frac{e_i}{\hat{\sigma} \sqrt{1 - h_{ii}}}.$$

Then D_i can be written compactly as

$$D_i = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}.$$

Step 4 — Identification with the F statistic.

Observe that D_i equals $1/p$ times the F -statistic for testing the hypothesis $H_0: \beta = \hat{\beta}_{(i)}$ at the full-data estimate. More precisely,

$$p D_i \equiv F\text{-statistic for } H_0,$$

which has numerator degrees of freedom p and denominator degrees of freedom $n - p$. Hence $p D_i$ is referenced against an $F(p, n - p)$ distribution, and common practice uses the 50th percentile (approximately $F = 1$) as an informal cutoff.

Caveat. The F -distribution connection is an *approximation*, not an exact result in finite samples. The residual e_i and $\hat{\sigma}^2$ share the same residual vector, so they are not independent; exact independence (required for a true F ratio) holds only asymptotically. The $F(p, n - p)$ percentiles therefore provide a natural *scale* for assessing influence rather than a formal significance level.